EC601 / SPRINT 1

Fidelity-Aware Low-Light Image Enhancement

A prototype for enhancement and spatial reliability assessment





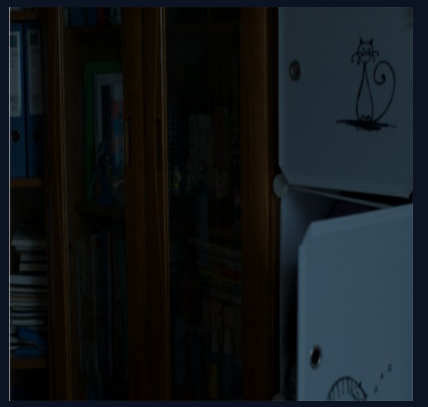
TEAM Fidelity-Aware LLIE • MEMBER Shangde Liu

Enhance the image. Question the result.

Primary user / photographers editing underexposed images

OUR MISSION

Help photographers enhance low-light images while indicating which regions of the result may be less reliable.



INPUT



ENHANCED

THE RESEARCH QUESTION

Can a spatial indicator flag regions that deserve closer inspection?

No claim of pixel-level truth: the indicator is an experimental proxy.

Five stories. One end-to-end workflow.

Sprint 1 touched all five stories; three are implemented in the initial prototype.

01	**Load paired low-light images** Input and reference load successfully	DONE
02	**Enhance a low-light image** Gamma works; stronger LLIE pending	IN PROGRESS
03	**Generate output sensitivity** Controlled input perturbation	DONE
04	**Visualize image and maps** Aligned six-panel comparison	DONE
05	**Validate indicator vs. error** One-image pilot; multi-image next	IN PROGRESS

A small, reproducible local pipeline.

Each tool has a clear purpose; no paid API or cloud dependency.

PYTHON

Pipeline and evaluation

NUMPY

Pixel-level arrays + statistics

PILLOW

Image loading and saving

MATPLOTLIB

Maps and comparisons

LOL DATASET

Paired low / reference images

GITHUB + VS CODE

Version control and workflow

INITIAL ENHANCEMENT BASELINE Gamma correction (γ = 0.4) • not a trained LLIE model

The prototype runs end to end.

Actual LOL evaluation image • gamma baseline • controlled Gaussian perturbation



LOW-LIGHT INPUT



PERTURBED INPUT



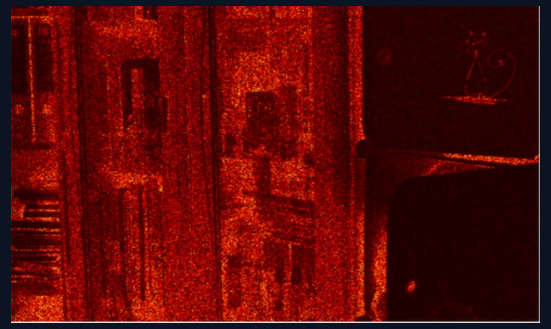
GAMMA ENHANCEMENT



NORMAL-LIGHT REFERENCE



REFERENCE ERROR MAP



OUTPUT SENSITIVITY MAP

ONE-IMAGE PILOT

−0.564

SENSITIVITY ↔ ERROR

+0.569

SENSITIVITY ↔ DARKNESS

−0.969

ERROR ↔ DARKNESS

3 focal papers: SID · Retinex-Net · MR. Illuminate

Pilot only — one image, one perturbation realization; not a validated reliability result.

A useful-looking map can measure the wrong thing.

The first experiment challenged our original assumption.

INITIAL HYPOTHESIS

Higher output sensitivity may indicate larger enhancement error.

OBSERVATION

Sensitivity and error were negatively correlated in the pilot; brightness is a likely confound.

WHAT CHANGED

A correlation is not enough. First ask whether the reference-error target is valid — then test whether sensitivity adds information beyond brightness.

Important: r = −0.969 comes from one image; it is a warning sign, not a dataset-wide conclusion.

From a working prototype to a falsifiable test.

Next two weeks / one reproducible evaluation and a clear pivot decision

01 **EVALUATE**
Run on multiple LOL image pairs

02 **COMPARE**
Sensitivity vs. darkness; two perturbation strengths

03 **STRENGTHEN**
Try one reproducible pretrained LLIE model

CURRENT BLOCKER

Choosing a robust LLIE baseline and checking whether reference error is a meaningful target.

SPRINT 2 DEMO SENTENCE

Show a reproducible multi-image evaluation and decide whether sensitivity adds value beyond brightness.